Primakoff axion production (γ ↔ a) implemented in OSIRIS

This note documents the Primakoff Monte-Carlo module added to the OSIRIS QED PIC-MC branch. It describes the physical model, how local conversion probabilities are computed, how macro-particle weights are handled (including the Poisson emission scheme and depletion), and where the module plugs into the OSIRIS timestep.

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1. Physical process and modeling assumptions

Primakoff conversion is the process γ + (Z) -> a + (Z), i.e. a photon converts into an axion in the Coulomb field of ions. In the implementation, ions act as a static scattering background: we compute a conversion rate for each photon using the local ion density and a screened Primakoff cross section. The emitted axion inherits the parent photon momentum (same direction and |p|), so the conversion is energy-conserving event-by-event within the macro-particle picture.

The inverse process (a -> γ) is implemented with the same rate model and Monte-Carlo bookkeeping so that the forward and inverse channels are algorithmic mirrors.

2. Integration into the OSIRIS timestep

Primakoff conversion is executed inside the QED advance loop (per QED group) during advance_deposit_qed. After QED-group merging, the code optionally performs inverse Primakoff (a -> γ) and then forward Primakoff (γ -> a), before advancing photons and axions with their usual pushers.

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Call ordering inside advance_deposit_qed (per QED group):
  1) if if_inverse:   primakoff_axion_to_photon(axions, photons, dt)
  2) if if_primakoff: primakoff_photon_to_axion(photons, axions, dt)
  3) advance photons, then advance axions, then advance charged species (superclass)
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Local ion density is provided to the Primakoff routines through a dedicated VDF (rho_ion) stored in the photon species. This rho_ion VDF is allocated during photon initialization when Primakoff is enabled, and is filled by depositing the selected ion species (same coupling convention as Bethe-Heitler).

3. Input interface and runtime wiring

Runtime parameters are read from the namelist group nl_qed_axion_primakoff. The photons species owns an axion_primakoff_info structure that stores these parameters, a pointer to the chosen ion species, and the rho_ion VDF.

Parameter	Meaning	Default / type
if_primakoff	Enable forward Primakoff conversion γ -> a for this QED group.	logical, default .false.
if_inverse	Enable inverse Primakoff conversion a -> γ.	logical, default .false.
g_agamma	Axion-photon coupling g_{aγ} in GeV^{-1}.	real, default 0
Te_keV	Plasma temperature used for screening κ; also default photon_Te_keV when Planck init is on.	real, default 0
use_screening	Use screened Coulomb field (κ > 0).	logical, default .true.
kappa_keV	If > 0, fixed κ in keV. If < 0, compute Debye κ from local densities.	real, default -1
Z_ion	Ion charge state Z used in σ_Prim(ω,κ).	integer, default 1
assume_neutral	If true, infer n_e from ions for Debye κ.	logical, default .true.
ne_factor	Multiplier in n_e = ne_factor * Z * n_i when assume_neutral is true.	real, default 1
include_ee	If true, include electron term in n_eff = n_e + Σ(Z^2 n_i) for κ.	logical, default .true.
w_axion	Nominal macro-axion weight used by Poisson emission.	real, default 1
lambda_cap	Cap on Poisson mean λ_eff; weights are scaled when cap triggers (unbiased).	real, default 20
photon_init	Photon momentum initialization: 'none' or 'planck' (applied once).	string, default 'none'
photon_Te_keV	Temperature for Planck photon_init (keV). If <= 0, defaults to Te_keV.	real, default -1
photon_quiet_pairs	If true, initialize photons in ±p pairs to reduce net momentum.	logical, default .true.
photon_nmax	Truncation parameter for Planck sampling (max 200).	integer, default 200
debug	Print optional debug statements.	logical, default .false.
i_ion	Ion species selector (same convention as Bethe-Heitler).	int array, default 1

Ion selection follows the established Bethe-Heitler convention: the array i_ion selects which ion species in the QED group is used as the Primakoff scattering target.

4. Local density and screening model

For each photon, the code gathers the local ion charge density from rho_ion using a nearest-grid-point (NGP) lookup. The deposited value is interpreted as a normalized charge density ρ_norm in OSIRIS units, from which the physical ion number density is reconstructed as n_i = |ρ_norm| n0 / Z, where n0 is the OSIRIS reference density for the QED group.

Screening enters through the parameter κ (kappa) that regulates Coulomb-field momentum transfer. If kappa_keV > 0 is provided, that value is used directly. Otherwise κ is computed from a Debye-Hückel model in natural units:

$$n_{eff} = n_e + \Sigma\,(Z^2\, n_i)$$
$$\kappa^2 = 4\pi\,\alpha\, n_{eff} / T$$

Implementation details:
- n_e is inferred from ions if assume_neutral is true:
  $n_e = ne_factor * Z * n_i$
- densities are carried in cm^-3 and converted to keV^3 using (ħc)^3
- κ is returned in keV

5. Primakoff cross section and per-photon rate

Given photon energy ω and screening scale κ, the total screened Primakoff cross section is evaluated as

Let $x = 4\,\omega^2 / \kappa^2$.

$$F(x) = (1 + 1/x)\ln(1 + x) - 1$$

$$\sigma_{Prim}(\omega, \kappa) = (g_{a\gamma}^2\,\alpha\, Z^2 / 8) * F(x) \quad \text{[natural units]}$$
$$= \sigma_{nat} * (\hbar c)^2 \quad \text{[converted to cm^2]}$$

The per-photon conversion rate is then

$$\Gamma(\omega) = n_i * \sigma_{Prim}(\omega, \kappa) * c \quad [s^{-1}]$$

The OSIRIS timestep is converted to physical seconds via dt_phys = dt / ω_p0, where dt is the normalized OSIRIS timestep and ω_p0 is the reference plasma frequency for the QED group. The conversion probability over one step uses the exponential survival form:

$$p_{conv} = 1 - \exp(-\Gamma\, dt_{phys}) \quad (\text{with } p_{conv} \approx \Gamma\, dt_{phys} \text{ when } \Gamma\, dt_{phys} \text{ is very small})$$

6. Monte-Carlo emission, weights, and photon depletion

Photons are represented by macro-particles with a positive weight q_ph (the number of physical photons represented). Axions are emitted as macro-particles of nominal weight w_axion. For each photon macro-particle, we convert the physical conversion probability into an expected number of emitted macro-axions:

λ = (q_ph * p_conv) / w_axion

k ~ Poisson(λ_eff), where λ_eff = min(λ, lambda_cap)

If λ is capped, emitted weights are scaled to keep the mean unbiased:
  scale = λ / λ_eff
  q_emit = w_axion * scale

For each of the k emissions, the code creates a new axion particle at the photon location and with the photon momentum vector. To enforce photon depletion, the total emitted axion weight q_total is subtracted from q_ph, and the photon macro weight is updated in-place. If q_total would exceed q_ph, the routine truncates the integer part and optionally emits one remainder-weight particle.

Momentum/energy inheritance (per event):
  p_a = p_γ (same 3-vector in OSIRIS units)
  ω_keV = |p_γ| * (m_e c^2) with m_e c^2 = 511 keV

Photon depletion:
  q_ph <- max(0, q_ph - q_total)

7. Optional Planck initialization of the photon bath

To support thermal Primakoff tests, the module can resample photon momenta exactly once at runtime when photon_init='planck'. This occurs at the start of the forward Primakoff routine and is gated by a photon_init_done flag so it is not repeated. If restarting from checkpoints and you do not want to reinitialize photons, set photon_init='none' in the restart deck.

Sampling strategy. For w = ω/T, the target number-spectrum pdf is f(w) ∝ w^2/(exp(w) - 1). The implementation uses a Bose-Einstein series method:

Algorithm:
  1) Draw an integer n with probability P(n) ∝ 1/n^3 (truncated at photon_nmax).
  2) Draw u1,u2,u3 ~ U(0,1) and form x_i = -ln(u_i).
  3) Set w = (x1 + x2 + x3)/n.

Directions are assigned isotropically. If photon_quiet_pairs is true, photons are initialized in ±p pairs to suppress net momentum and reduce shot noise.

8. Inverse Primakoff (a -> γ)

The inverse of the above is implemented in primakoff_axion_to_photon. The routine loops over axion macro-particles, computes the same local ion density and screening scale, forms Γ and p_conv, samples k ~ Poisson(λ_eff) with the same cap-and-rescale method, creates photons with the parent axion momentum, and depletes the axion macro weight by the emitted total weight.

References

G. G. Raffelt, Phys. Rev. D 33, 897 (1986). (Reference for screened Primakoff-style rate expressions and screening conventions.)

OSIRIS QED PIC-MC infrastructure (QED-group advance, photon species, Bethe-Heitler ion-density deposition) provides the integration scaffolding.